

Optimizing Trading Strategies with an LSTM and a Dynamic Attention Mechanism

Nikola Jarić · September 2024 · Code: github.com/njaric03/crypto-lstm-predictor

Summary

This report describes a short-term price direction model for cryptocurrency markets and the trading strategy built on top of it. The model is an LSTM whose attention weights are generated dynamically from current volatility and trading volume rather than learned as fixed parameters, so the amount of history the model leans on shifts with market conditions. Inputs are standard technical indicators compressed with PCA. Backtested on BTC/USDT from January 2023 to May 2024, the strategy returned 335.7% against 197.0% for buy-and-hold, with the clearest advantage appearing in falling months. Section 4 sets out why that number should be read as an upper bound.

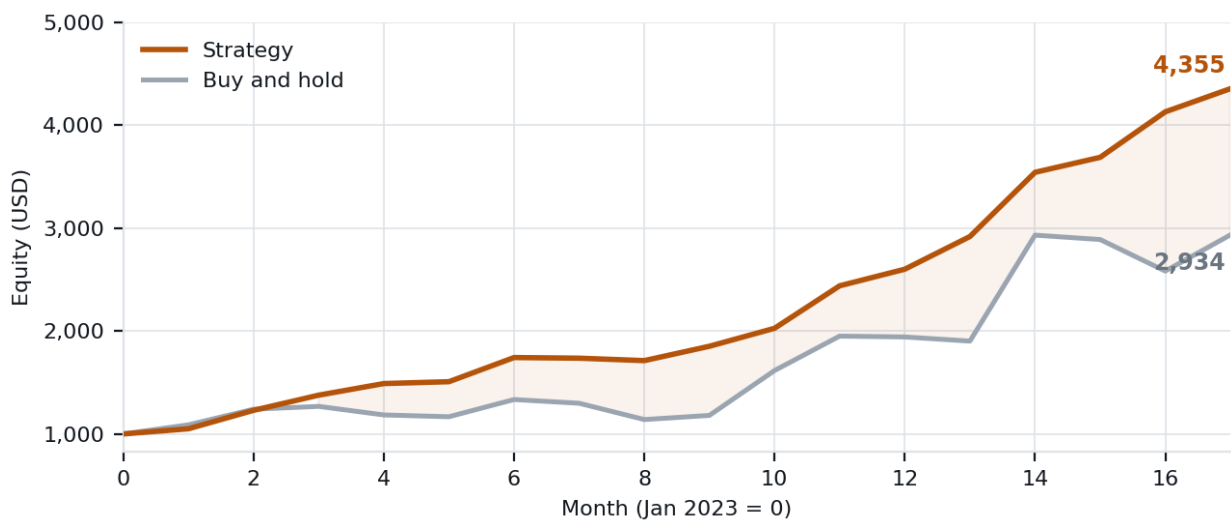


Figure 1. Equity from a 1,000 USD starting position, compounded from monthly returns. Fees and slippage are not deducted; see section 4.

1. Model architecture

LSTMs handle long-range dependence in sequences through gated cell state, which makes them a reasonable default for financial time series. The addition here is the attention layer. Static attention applies fixed weights across the input window; this implementation derives the weights at inference time from two market signals, volatility and volume, and multiplies them into the LSTM output before the softmax. In a calm market the model spreads attention broadly; in a volatile one it concentrates on recent steps.

```
class DynamicAttention(nn.Module):
    def __init__(self, hidden_dim):
        super().__init__()
        self.feature_layer = nn.Linear(2, hidden_dim, bias=False)
        self.attention = nn.Linear(hidden_dim, 1, bias=False)

    def forward(self, lstm_out, volatility, volume):
        features = torch.cat((volatility.unsqueeze(-1),
                              volume.unsqueeze(-1)), dim=-1)
        dynamic_weights = torch.tanh(self.feature_layer(features))
        attention_weights = torch.softmax(
            self.attention(lstm_out * dynamic_weights).squeeze(-1), dim=1)
        context = torch.sum(attention_weights.unsqueeze(-1) * lstm_out, dim=1)
        return context, attention_weights
```

The context vector then passes through a stack of fully connected layers with LeakyReLU, dropout and batch normalisation, narrowing from *hidden_dim* to *hidden_dim/4* before the classification head. Output is a

three-way trend label: up, down, or neutral.

2. Features and dimensionality reduction

Inputs are price and volume history plus a set of technical indicators: SMA and EMA, RSI, the stochastic oscillator, Bollinger Bands, MACD, ATR, and the Ichimoku Cloud components. These overlap heavily by construction, so PCA is applied before the sequence reaches the LSTM in order to decorrelate the feature set, suppress noise, and reduce the parameter count the model has to fit.

Configuration	Training time (s)	Test loss
PCA, 32 hidden dimensions	4,172.34	1.1040
No PCA, 64 hidden dimensions	4,293.69	1.1144

Table 1. PCA at half the hidden size trains 3% faster and reaches a 0.5% lower cross-entropy test loss.

The gain is small but the direction matters: halving the hidden dimension cost nothing in accuracy, which suggests the uncompressed feature set carried redundancy rather than signal.

3. Backtest results

The strategy was evaluated on 17 months of BTC/USDT data, January 2023 through May 2024, starting from 1,000 USD. Final equity was 4,356.98 USD, a 335.7% return against 197.0% for holding the asset over the same window, across 13,243 executed trades.

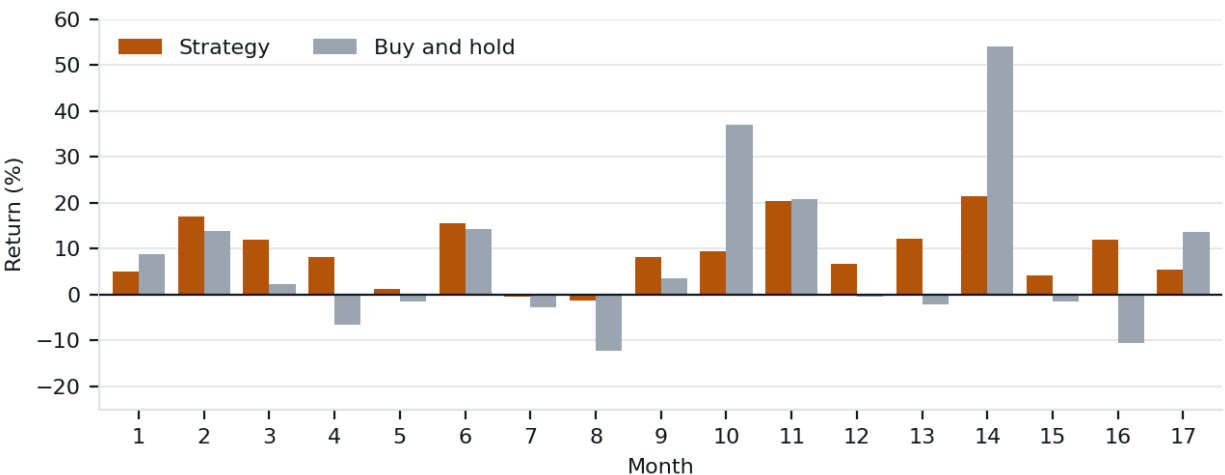


Figure 2. Monthly return, strategy versus buy-and-hold.

The headline return is the least interesting number here. What holds up better is the behaviour in bad months. In month 8 the market fell 12.20% while the strategy lost 1.36%. In month 4 the market fell 6.57% and the strategy returned 8.20%; in month 16 the market fell 10.61% and the strategy returned 11.99%. The strategy also underperforms badly in the two strongest bull months, 10 and 14, where holding returned 36.94% and 54.14% against 9.43% and 21.38%. The pattern is consistent with a model that trims exposure into weakness and therefore gives up upside in sharp rallies.

4. Limitations

Three caveats are worth stating plainly, because they bound how the result above should be read.

- **Transaction costs are not deducted.** 13,243 trades over 17 months is a high turnover strategy, and at typical exchange taker fees the accumulated cost is material. The 335.7% figure is a gross backtest return, not a realised profit.

- **Slippage and liquidity are not modelled.** Fills are assumed at the observed price, which flatters any strategy trading at this frequency.
- **A single asset over a single regime.** The window covers one recovery cycle in BTC/USDT. Nothing here demonstrates that the model generalises to other assets or to a different volatility regime.

The correct reading is that dynamic attention improved the model's response to changing market conditions relative to the static baseline, and that the drawdown behaviour is worth further work. Live profitability is not established by this report.

5. Further work

The obvious next steps are a transformer baseline for comparison, a cost model applied to the backtest so net returns can be reported, sentiment features alongside the price-derived ones, and evaluation across several assets and a longer window. Chart pattern recognition and macroeconomic inputs such as rates and inflation are a longer-term direction.